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Original Study

Effect of an Aerobic Dancing Program on Sleep Quality for Older Adults With Mild Cognitive Impairment and Poor Sleep: A Randomized Controlled Trial

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Keywords:
Exercise
sleep quality
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A B S T R A C T

Objective: To investigate the effects of an aerobic dancing program on sleep quality of older adults with mild cognitive impairment (MCI) and poor sleep.
Design: The study adopted a single-blinded randomized controlled trial design.
Settings and Participants: Community-dwelling adults older than 60 years screened with the coexistence of MCI and poor sleep were included. A total of 89 eligible participants were randomly assigned to receive either the aerobic dancing program or the health education attention placebo control group.
Methods: The primary outcome was sleep quality, which was measured by the Pittsburgh Sleep Quality Index. The second outcome was cognitive function, which was assessed by the Montreal Cognitive Assessment. Analysis followed the intention-to-treat principle. Generalized estimating equation was adopted to compare the changes in outcomes over the pre-post test periods between groups.
Results: The average adherence rate of the exercise program was 73.1%. No adverse events occurred during the study period. Participants in the intervention group had a significantly greater improvement in terms of general sleep quality (β, −1.74; 95% CI, −3.41 to −0.08; $P = .04$), sleep latency (β, −0.19; 95% CI, −0.37 to −0.02; $P = .03$), sleep duration (β, 0.51; 95% CI, 0.24 to 0.99; $P = .04$), sleep efficiency (β, 0.11; 95% CI, 0.07 to 1.12; $P = .04$), and cognitive function (β, 1.64; 95% CI, 0.43 to 2.86; $P = .008$) compared with the control group over the pre-post test periods.
Conclusions and Implications: This study demonstrated that a 16-week aerobic dancing program is feasible and effective in improving sleep quality and cognitive function among older adults with coexistence of MCI and poor sleep. The feasibility and effectiveness of this exercise program promoted the need to increase its application against deteriorating poor sleep and cognitive decline in older adults with MCI.

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Older adults with mild cognitive impairment (MCI) are at a high risk for progressing to dementia.[1] Approximately 10% to 15% of older adults with MCI developed dementia annually, whereas the overall annual conversion rate to dementia is approximately 1% to 3% in general older adults.[2] MCI is prevalent among older adults with an estimated global prevalence range of 16% to 22.2%.[3-5] Because there is no definite cure for dementia, greater efforts are now focused on the modifiable risk factors that may lead to cognitive deterioration at this stage.

Poor sleep quality is receiving increasing attention as a prominent risk factor for cognitive impairment.[6] Indeed, poor sleep at nighttime leads to daytime sleepiness and fatigue, which leads to decreased alertness, withdrawal from social activities, and increasing variability in cognitive performance.[7,8] Longitudinal study evidence indicated that poor sleep quality almost doubles the risk of cognitive impairment and reduces the chance of reversion to normal cognition by 31% in older adults with MCI.[9] Poor sleep is commonly reported by individuals with MCI, with a prevalence estimate ranging from 18.5% to 75.5%.[10-12] The high prevalence and cognitive declining effects of poor sleep urges the need to identify effective interventions targeting this situation in MCI.

Currently, the interventions targeting sleep quality include pharmacological and nonpharmacological measures.[13,14] Pharmacotherapy

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is recommended for short-term use only because of commonly occurring side effects, such as rebound insomnia and falls.[15] Of particular concern is the potential to exacerbate cognitive impairment in older adults.[16,17] The first-line nonpharmacological therapy for poor sleep is cognitive behavioral therapy.[18] However, cognitive behavioral therapy remains limited and unavailable to some extent, owing to resource unavailability and financial cost. Exercise as a low-risk, low-cost lifestyle intervention is recognized as an alternative nonpharmacological therapy for poor sleep.[19]

There is compelling evidence to indicate that exercise programs positively affect sleep outcomes in generally healthy older adults.[20] Results of several systematic reviews of intervention studies have shown consistent benefits of exercise on total sleep time, sleep efficiency, sleep onset latency, and overall sleep quality.[21-23] Indeed, the results of biological studies indicate that the beneficial effects of regular exercise on sleep may be explained by multiple pathways with the interaction of circadian rhythm, immune, thermoregulatory, metabolic, vascular, and mood effects.[24] Among all the exercise modalities, aerobic exercise (ie, walking, jogging, dancing) is more studied because it is more readily available than other modalities, such as resistance training. Further, a recent randomized controlled trial compared the impact of aerobic and resistance exercise training on sleep quality and inflammatory markers among 60 adults with chronic primary insomnia and found that aerobic exercise is more appropriate than resistance training in modulation of inflammation and sleep quality.[25]

Among older adults with MCI, there is still a lack of investigations on whether exercise programs benefit sleep quality among older adults with MCI and poor sleep. Given the solid research evidence on the superior effects of aerobic exercise on sleep outcomes in the general older population, this study therefore aimed to fulfill the gap by investigating the feasibility and effects of an aerobic dancing program on sleep quality of older adults with both MCI and poor sleep. We hypothesized that the 16-week exercise program would be effective in improving the sleep quality of older adults with MCI and poor sleep.

Materials and Methods

Design and Sample

This was a single-blinded randomized controlled trial (registration number: CUHK_CCRB00531), and ethical approval was obtained from The Chinese University of Hong Kong (No. 2016.287). The random sequence was generated by computer software. Random numbers were kept in opaque and sealed envelopes, which were prepared by a research assistant who did not participate in data collection. All participants provided written informed consent.

A total of 89 participants were recruited from 2 community health care centers (CHCs) in downtown Hangzhou City, China. In China, CHCs are set up by the public sectors to provide basic medical and public health services. The main medical services provided by CHCs include health consulting, diagnosis, and treatment for the common and frequently occurring chronic diseases. Inclusion criteria were (1) community-dwelling people aged 60 and older, (2) who scored between 19 and 26 of 30 on the Montreal Cognitive Assessment (MoCA); this cutoff score correctly identified more than 90% of a large sample of MCI,[26] appropriate permission has been obtained from the copyright holders of this scale, and (3) perceived poor sleep quality indicated by a score >5 of 21 on the Pittsburgh Sleep Quality Index (PSQI).[27] Exclusion criteria were (1) conditions that were contraindicated for exercise according to the American College of Sports Medicine (ACSM), (2) neurological disorders (eg, stroke, brain injury), which impairs cognition greatly, or (3) regularly involved in moderate or vigorous-intensity exercises of 150 minutes per week.

Sample Size

Sample size was calculated by power analysis. According to a systematic review of 66 experimental studies that examined the effects of exercise intervention on general sleep quality among adults,[21] the effect size of regular exercise interventions on sleep quality was 0.7 (Cohen's d). Accordingly, the present study aimed to detect a moderate effect size of 0.7 on the primary sleep quality outcome. Using the Gpower program,[28] it is estimated that a sample size of 40 participants per arm would give the research project 80% power at a 5% level of significance to detect the targeted effect size, allowing for a 10% attrition.

Intervention

Participants in the experimental group received a 16-week aerobic dancing program. This exercise program consisted of three 60-minute group training sessions per week in a group-based format (20 participants per group). The exercise program was designed by a team comprising a nurse academician, a physiotherapist, and exercise physiologist following the recommendations of the ACSM position stand on exercise and physical activity for older individuals.[29,30]

The 1-hour training program started with 10-minute warm-up session, after that a 40-minute aerobic dancing session, and ended with a 10-minute cool-down session. Warm-up and cool-down exercise included walking and stationary stretching exercises for trunk and limb joints. The program involved dancing patterns of stepping up and down on a 10-cm stepping bench. Joyful music was integrated into the exercise regimen to facilitate adherence. The stepping dancing program was adopted because it is highly suitable for older people who have decreased joint flexibility and bone strength.[31] Meanwhile, various sets of entertaining upper limb movements resembling daily activities, such as washing the hands, were added to facilitate overall body movement and adherence.[30] The participants were instructed to adjust their movements to achieve exertion in moderate-intensity aerobic exercise, which was indicated by the feeling of "somewhat hard" at the Borg rating of 12 to 14.[32] The aerobic dancing program was conducted by 2 nurses, one serving as an instructor and the other as a facilitator. Vital signs and contradictory symptoms were assessed before training.

Participants in the control intervention received 8 biweekly group-based health educational sessions (45 min/session) delivered by general practitioners in the CHCs. No information relating to physical exercise was included.

Measures

The primary outcome was general sleep quality, and the secondary outcome was cognitive function. All the outcomes were measured at baseline and at the end of the 16-week intervention period.

Sleep Quality

General sleep quality was measured by the PSQI, which consists of 24 questions regarding sleep latency, sleep duration, sleep efficiency, and so forth.[27] A cutoff score of 6 has a sensitivity of 89% and specificity of 86% in differentiating good and poor sleepers in Chinese.[33] The Cronbach's alpha of PSQI in this study was 0.736. The secondary sleep quality outcomes were measured by the subscales of PSQI, including sleep latency, sleep duration, sleep efficiency, sleep disturbance, and daytime dysfunction.

Cognitive Function

Cognitive function was measured by the MoCA using a paper-and-pencil approach. The MoCA contains tasks for episodic memory,

visuospatial ability, executive function, attention, language, and orientation to elicit a comprehensive overview of one's cognitive abilities.[26] The Chinese version of MoCA has good reliability with Cronbach's α being 0.836[34] and is more sensitive than the Mini-Mental State Examination in evaluating cognitive abilities in individuals with MCI.[35]

Feasibility of the Intervention

Attendance rate, retention rate, and occurrence of adverse events were used to evaluate whether the intervention program was feasible. Attendance rate was calculated as (number of sessions a participant attended/total number of sessions of the program) * 100%. Retention rate was calculated as (number of participants who had completed the program/number of recruited participants) * 100%. The program would be perceived as feasible if the retention rate was at or above 80% and if the attendance rate was at or above 75%. The occurrence of adverse events was recorded during each session.

Statistical Analyses

All data were analyzed using SPSS version 22.0 (SPSS Inc.). The χ^2 and *t*-tests were adopted for checking for any significant difference in demographic and outcome variables between 2 study arms at baseline. To detect the intervention effects, the generalized estimating equation (GEE) model was used to analyze the group and time differences of outcome variables. The GEE model is commonly used in longitudinal studies for data analysis as it accounts for possible intracorrelations in repeated measures over time and can explore group differences with adjustment for covariates.[36] Another advantage of adoption of the GEE model is its ability to fit different types of data with the use of an appropriate link function.[37] The GEE model is an optimal choice for data analysis in longitudinal intervention studies, as the GEE model allows the use of partial information relating to participants who dropped out across the study period to contribute to estimating the model parameters when the data are missing at random.[36] The intention-to-treat principle was followed. A significance level of .05 was set for all analyses.

Results

Sample Characteristics

As shown in the Consolidated Standards of Reporting Trials (CONSORT) flow diagram (Figure 1), 512 adults older than 60

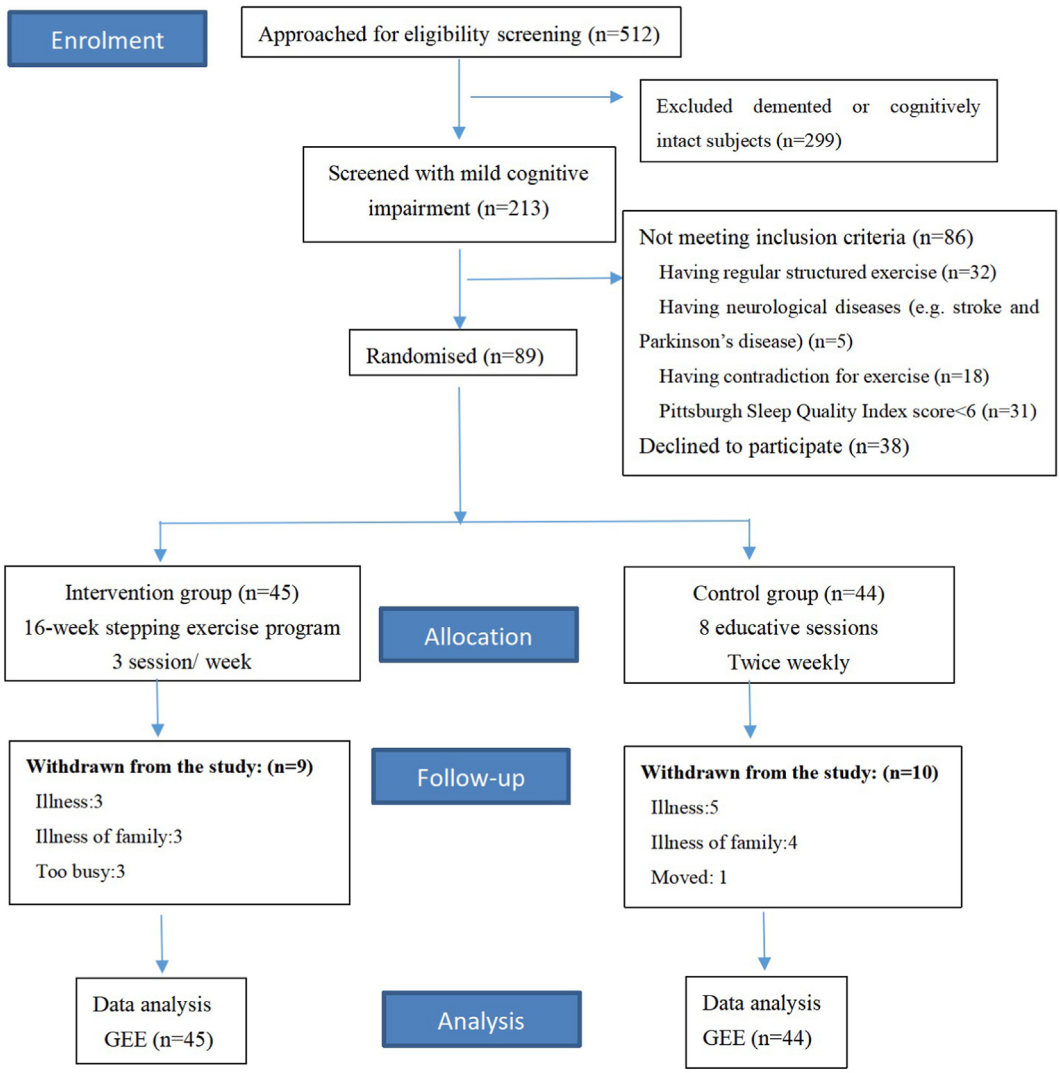

Fig. 1. The CONSORT diagram.

Table 1
The Characteristics of the Respondents (N = 89)

Demographic Characteristics	Overall (N = 89)	Intervention (n = 45)	Control (n = 44)	*P*
Age, y	75.97(6.31)	76.71 ± 5.96	75.20 ± 6.63	.26
Gender				.42
Female	90	36 (80.0%)	32 (70.0%)	
Male	30	9 (20.0%)	12 (30.0%)	
Marital status				.59
Married	81	31 (65.0%)	30 (70.0%)	
Single	39	14 (35.0%)	14 (30.0%)	
Education level				.73
Less than middle school	51	20 (46.7%)	14 (38.3%)	
Middle school	42	15 (35.0%)	19 (35.0%)	
More than middle school	27	10 (18.3%)	11 (26.7%)	
Residence				.15
Living alone	32	15 (33.3%)	7 (20.0%)	
Living with spouse	68	24 (51.7%)	28 (61.7%)	
Living with children only	20	6 (15.0%)	9 (18.3%)	
Monthly income (CNY)				.45
Less than 4000	76	28 (66.7%)	29 (60%)	
4000 and more	44	17 (33.3%)	15 (40%)	

Note: 1 US Dollar = 7 CNY.

were approached for eligibility screening. Among the 127 eligible participants, 89 consented to participate. The baseline characteristics of the participants in 2 groups are shown in Table 1. No significant differences were found between 2 groups in terms of demographic characteristics, baseline outcomes, and cognitive function.

Feasibility

During the 16-week study period, 19 participants discontinued (attrition rate: 21.3%) because of illness (n = 8) or unanticipated family duties (n = 11). Following the intention-to-treat principle, post-test data were also collected from these participants with their consent. The average adherence rate of the exercise program was 73.1%. No adverse events occurred during the study period.

Effects of the Aerobic Dancing Program on Sleep Quality

Table 2 presents the baseline and post-test PSQI scores of both groups. Participants in the intervention group had significantly greater reduction in their PSQI total score compared with the control group across the pre-test and post-test periods (β, −1.74; 95% CI, −3.41 to −0.08; P = .04), which indicates a significant improvement in general sleep quality. As for the secondary sleep outcomes, the GEE results were significant for sleep duration (β, 0.51; 95% CI, 0.24 to 0.99; P = .04), sleep latency (β, −0.19; 95% CI, −0.37 to −0.02; P = .03), and sleep efficiency (β, 0.11; 95% CI, 0.07 to 1.12; P = .04). The GEE results were not significant for sleep disturbance (β, −0.13; 95% CI, −0.58 to −0.33, P = .59) and daytime dysfunction (β, 0.06; 95% CI, −0.17 to −0.19; P = .94).

Effects of the Aerobic Dancing Program on Cognitive Function

Table 2 presents the baseline and post-test MoCA scores of both groups. Participants in the intervention group had significantly greater improvement in their MoCA score compared with the control group across the pre-test and post-test periods (β, 1.64; 95% CI, 0.43 to 2.86; P = .008), which indicates a significant improvement in cognitive function.

Discussion

This study aimed to investigate whether a 16-week aerobic dancing program would be feasible and effective in improving sleep quality of older adults with coexistence of MCI and poor sleep. The findings of the study showed that the aerobic dancing program significantly improved sleep quality for older adults with MCI and poor sleep. This exercise intervention also effectively reduces the effect of poor sleep on the cognitive function of older adults with MCI. Notably, the attrition rate and adherence rate of the exercise intervention in this study was acceptable, although evidence has documented challenges in retaining patients in long-term physical exercise interventions.[38]

Consistent evidence pointed out that physical exercise is effective in promoting sleep quality among general older adults[22]; our study is among the very few studies that applied an exercise intervention in older adults with MCI and poor sleep. In the present study, the effect size of the exercise program on general sleep quality was 1.4, which is larger than the average effect size (0.7) generated by a systematic review of 66 experimental studies that examined the effects of exercise on general sleep quality among general adults.[21] Both difference in participant characteristics and intervention characteristics may explain the greater effect size in our study. Unlike the systematic review that included general adults as participants, we included older adults who had coexistence of poor sleep and MCI, who may be more respondent to sleep interventions. In terms of intervention design, this aerobic dancing program is developed according to the recommendations of the ACSM, so that participants would have regular aerobic exercise at moderate intensity 3 times a week. By teaching the participants how to use the Borg's scale to pace their activity, the instructor found that they had achieved moderate-intensity exercise output most of the time during the training phase. Exercise intensity monitoring has seldom been mentioned in previous studies. The regular moderate-intensity aerobic exercise would result in more sustainable vascular and mood effects,[24] which favors sleep quality. Only 1 previous study was found to apply a 20-week aerobic walking program in older adults with MCI and poor sleep,[39] which reported an approximate effect size of 3. However, the authors admitted that this study is not vigorous in measurement, because of no assessor blinding and a blank control, which makes this effect size questionable.

By exploring the effects of the exercise intervention on PSQI subscale score, our study indicated that this aerobic exercise program also reduced sleep latency, improved sleep duration, and enhanced sleep efficiency. This is consistent with the findings reported by previous studies using objective sleep measures, such as polysomnography, to examine the effects of exercise interventions on sleep outcomes in general older adults.[40,41]

Poor sleep quality is highly prevalent among older adults with MCI. Our previous study reported that the prevalence of self-perceived poor sleep is 70.1% in MCI over the 56.5% in the cognitive healthy controls.[12] Mounting evidence points to a potential connection between sleep and cognitive function. Poor sleep greatly interferes with the daily activities of older adults with MCI and significantly predicts further cognitive deterioration. Indeed, research indicated that poor sleep escalates the production of amyloid-β oligomer and impairs metabolic activities in the prefrontal cortex.[42] Through the positive effects of exercise training

Table 2
GEE Model for the Comparison of Outcome Variables Between Groups

Outcomes	Intervention Group	Control Group	Group Effect		Time Effect		Group* Time Effect		Effect Size (d)
			β (95% CI)	*P*	β (95% CI)	*P*	β (95% CI)	*P*	
PSQI-Overall									
Baseline	11.07 ± 2.65	10.66 ± 3.14	0.41	.49	0.20	.74	−1.74	.041	1.40
Post-test	9.52 ± 2.31	10.86 ± 3.15	(−0.76 to 1.58)		(−0.97 to 1.38)		(−3.41 to −0.08)		
PSQI-Sleep duration (Hour)									
Baseline	5.23 ± 0.78	5.34 ± 0.85	−0.11	.53	−0.05	.77	0.51	.039	0.67
Post-test	5.69 ± 0.78	5.29 ± 0.86	(−0.44 to 0.23)		(−0.39 to −0.21)		(0.24 to 0.99)		
PSQI-Sleep latency (Hour)									
Baseline	0.62 ± 0.32	0.57 ± 0.35	0.006	.39	0.003	.96	−0.19	.034	0.20
Post-test	0.42 ± 0.17	0.56 ± 0.34	(−0.07 to −0.18)		(−0.13 to 0.12)		(−0.37 to −0.02)		
PSQI-Sleep efficiency									
Baseline	0.67 ± 0.11	0.68 ± 0.12	−0.007	.78	−0.007	.79	0.11	.041	0.5
Post-test	0.77 ± 0.13	0.67 ± 0.13	(−0.06 to −0.04)		(−0.06 to 0.04)		(0.07 to 1.12)		
PSQI-Sleep disturbance									
Baseline	1.11 ± 0.38	1.07 ± 0.25	0.11	.51	0.03	.86	−0.13	.59	0.18
Post-test	1.12 ± 0.31	1.07 ± 0.26	(−0.21 to −0.42)		(−0.29 to −0.35)		(−0.58 to −0.33)		
PSQI-Daytime disfunction									
Baseline	1.29 ± 0.81	1.18 ± 0.82	0.04	.51	0.002	.98	0.06	.94	0.2
Post-test	1.19 ± 0.71	1.21 ± 0.77	(−0.08 to −0.17)		(−0.128 to −1.31)		(−0.17 to −0.19)		
MoCA									
Baseline	22.11 ± 2.03	22.02 ± 1.96	0.88	.83	−0.25	.56	1.64	.008	1.3
Post-test	23.50 ± 2.00	21.77 ± 2.27	(−0.76 to 0.94)		(−0.1.16 to 0.061)		(0.43 to 2.86)		

Note: group effect = the overall between-group changes; time effect = the within-subject changes over the evaluative endpoint in the control group; Group* time effect = additional score changes of intervention group compared with control group at post-test; β = regression coefficient.

on sleep quality, a favorable functional and physiological condition is created to improve the cognitive function of older adults with MCI, as indicated by the evidence generated from this study.

Our study has several limitations that may influence the interpretation of the results. First, because participants were recruited in CHCs through convenient sampling, the generalizability of the findings is limited. Second, only subjective sleep quality was assessed in this study; therefore, the findings did not provide evidence that suggests the benefits of this exercise program on objective sleep pattern in MCI. Third, the attrition rate was higher than anticipated and is greater than the 10% used in power calculations, which may diminish the study power. Finally, although attention placebo was adopted in this study as a control group, the control group did not receive attention equal to the intervention group, which may interfere with the interpretation of the difference in post-test study results between groups. Further research is needed to explore the preceding research gaps.

Conclusions and Implications

This study revealed that a 16-week aerobic dancing program is feasible and effective in improving sleep quality and cognitive function among older adults with MCI and poor sleep. The effectiveness and feasibility of this exercise program urges the need to increase its application against deteriorating poor sleep and cognitive decline in older adults with MCI. Lifestyle intervention is more affordable, accessible, and readily applicable to be integrated into older adults' daily lives. Health workers play key roles in promoting cognitive health, and professional training in this regard should be emphasized to promote the successful application of physical exercise in older adults with MCI. Furthermore, efforts should be devoted to ensuring the adherence of older adults with MCI with regular practice as a lifestyle routine. The group exercise format enhances the adherence of exercise interventions. Making good use of mobile health applications in facilitating such health initiatives should also be considered in future studies.

Disclosure

The authors declare no conflicts of interest.

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